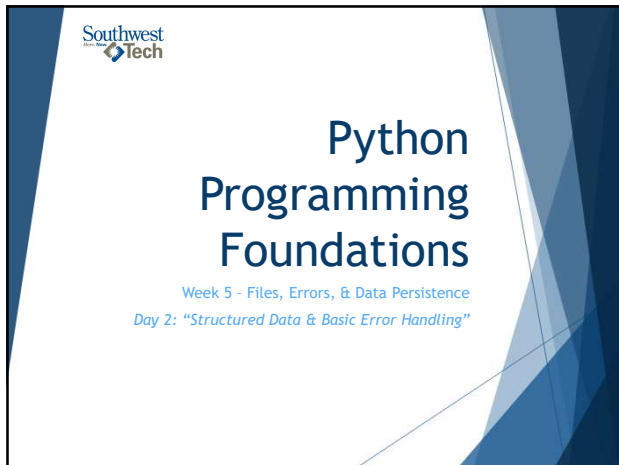
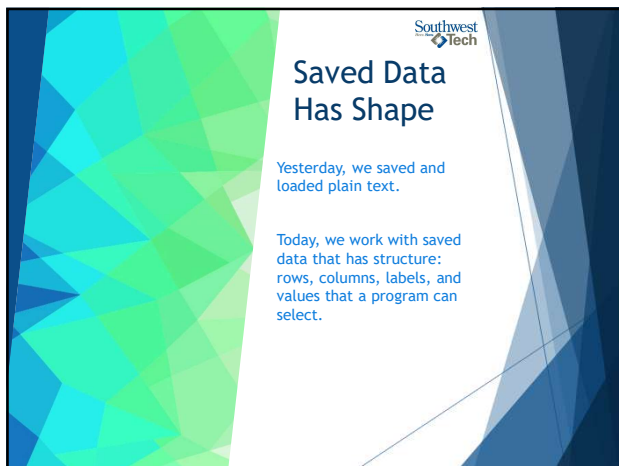
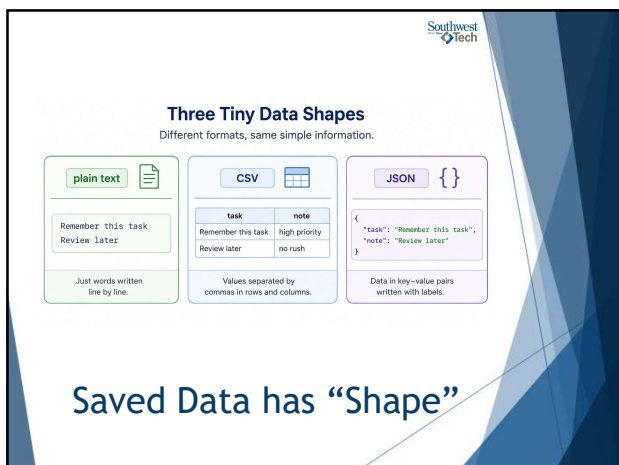




1



2



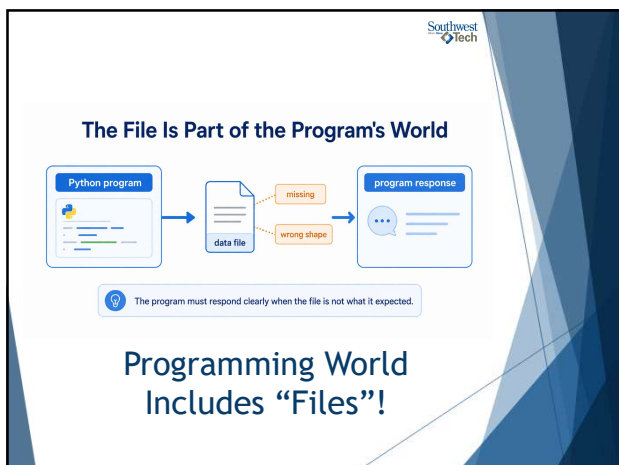
3



4



5



6

Southwest
Tech

What Counts as Success Today

Working baseline ->
compare structure ->
revise one meaningful thing ->
explain the improvement.

A useful revision should make the program:

clearer	easier to test	easier to change	easier to explain
---------	----------------	------------------	-------------------

7

Southwest
Tech

Reading Structured Data: A 5-Step Success Pattern

- 1 Identify file shape
- 2 read the file
- 3 select useful values
- 4 handle one likely problem
- 5 explain what the program did

Common File Shapes

plain text Remember this task Review later	CSV task_note Remember this task, high priority Review later, no rush	JSON <pre>{ "task": "Remember this task", "note": "Review later" }</pre>
--	---	---

The file shape affects the program's choices.

Today's Success

8

Southwest
Tech

What We Will Use Today

- ▶ CSV rows and columns
- ▶ JSON labels and values
- ▶ 'try' / 'except'
- ▶ selected-value output
- ▶ one clear error path

9

Southwest Tech

Today's Structured Data Tools

Simple tools to read, use, and handle data with confidence.



Toolbox

10

Southwest Tech

What We Will “Skip” For Now

- ▶ databases
- ▶ custom JSON encoding
- ▶ large data pipelines
- ▶ advanced exception design
- ▶ configuration systems



Today we read small structured files and respond to likely problems.

11

Southwest Tech

Save For Later



Parked For Later

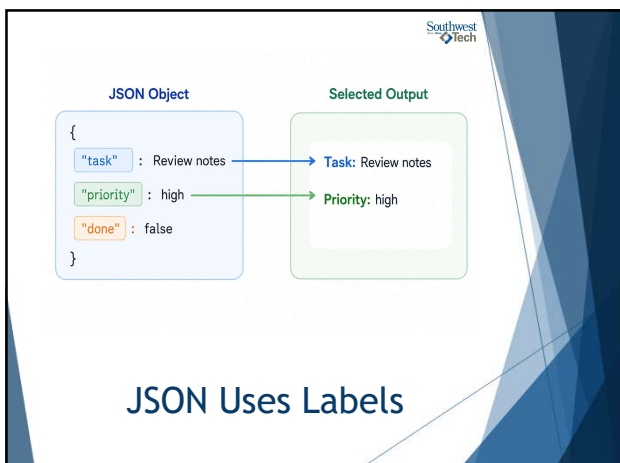
12



13



14



15



CSV Uses Rows And Columns

CSV data is usually organized like a table.

Each row represents one record, and each column represents one kind of value.

Think in terms of a spreadsheet

16



CSV Summary

	A	B	C
1	Date	Minutes	Topic
2	Mon	30	Python
3	Tue	20	Files
4	Wed	25	Practice

→ summarize →

Sessions: 3

Total minutes: 75

CSV Uses Rows & Cols

17




Loading Is Not The Finished Result

Reading a file is only the first step.

The program should:

- ▶ select,
- ▶ summarize,
- ▶ filter,
- ▶ or format information ...

so the output becomes useful.

18

Southwest Tech

Loaded Data Becomes Useful Output

Loaded data



```
id,task,done,order  
1,study files,false,1  
2,review notes,true,2  
3,practice,true,3  
... other fields ...
```

select /
summarize /
format

→

Useful output



Tasks completed: 3

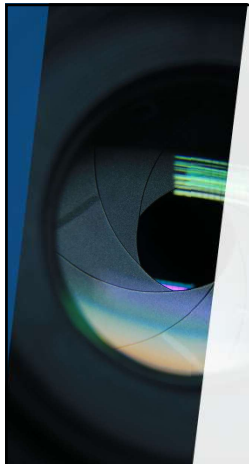


Next task: Review notes

💡 Reading the file is only the first step.

“Loading” is NOT the Finished Result

19



Demo #1

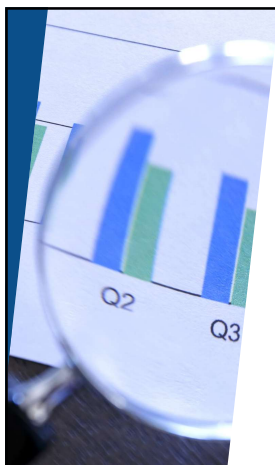
“Save And Load JSON Tasks”

20

Southwest Tech

What to Watch For

- ▶ List of tasks
- ▶ Labels inside each task
- ▶ Selected values when loaded



21

Southwest
Tech

JSON Tasks: Save, Load, Select

JSON Tasks

task: "study files"
done: false

task: "review notes"
done: true

load JSON

Selected Output

2 tasks loaded

Completed: review notes

Demo 1: Save & Load JSON Tasks

22

Demo #2

"Read A CSV Summary"

23

Southwest
Tech

What to Watch For

CSV data often supports
small reports.

Instead of printing every
row, the program can
count, total, filter, or
display selected records.

Q2

Q3

24

Southwest Tech

CSV Summary

Date	Minutes	Topic
Mon	30	Python
Tue	20	Files
Wed	25	Practice

→ summarize →

Sessions: **3**

Total minutes: **75**

Demo 2:
Read a CSV Summary

25

Southwest Tech

Not Every Failure Means The Same Thing

Different problems need different explanations.

- ▶ **missing file**: the program cannot find what it was told to read
- ▶ **invalid data**: the file exists, but the contents do not match expectations

26

Southwest Tech

Different Problems, Different Responses

File missing

Tell the user what file was not found

Invalid data

Tell the user the file contents need checking

Different problems need different, helpful messages.

Not Every Failure Means The Same Thing

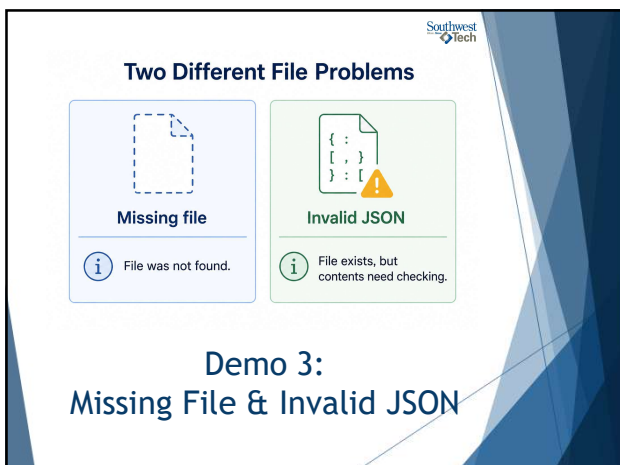
27



28



29



30

[illegible]

31

The diagram illustrates the difference between a raw dump and a useful output. It is divided into two main sections: 'Raw dump' and 'Useful output'.

Raw dump: This section shows a block of raw data, represented as a hexadecimal string: `a1b7,91c2,08f3,?,x9z {94aa|07bb|item=556|...} 92-11-ff / ?? / zz-88 !@#key_9|val:??|[]{} id=00391; t=15; f@!% .....` . This represents a raw dump of data.

Useful output: This section shows the same data processed into a structured format. It contains two items:

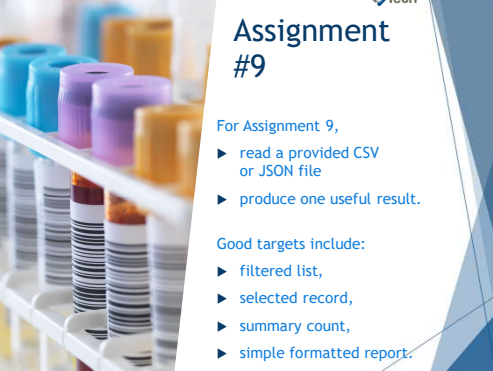
- Selected record:** Review notes
- Status:** complete

An arrow points from the 'Raw dump' section to the 'Useful output' section, indicating the transformation process.

Conclusion: A yellow box at the bottom states: "A useful result shows that the program understood the structure." This highlights the importance of structured output for understanding the data.

Common Failure: The diagram also highlights a common failure: "Raw Data Dumping". This is shown as a large, dark, and unstructured block of data, contrasting with the structured and useful output.

32





Assignment #9

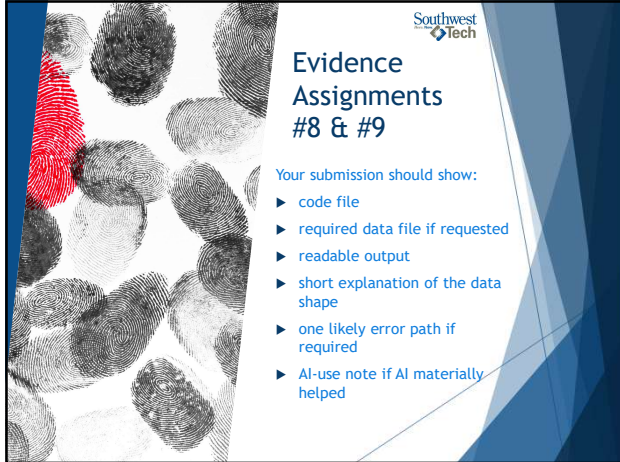
For Assignment 9,

- ▶ read a provided CSV or JSON file
- ▶ produce one useful result.

Good targets include:

- ▶ filtered list,
- ▶ selected record,
- ▶ summary count,
- ▶ simple formatted report.

33



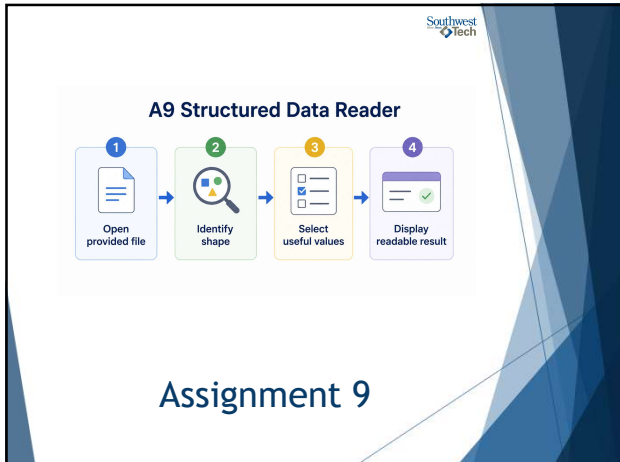
Southwest
Tech

Evidence Assignments #8 & #9

Your submission should show:

- ▶ code file
- ▶ required data file if requested
- ▶ readable output
- ▶ short explanation of the data shape
- ▶ one likely error path if required
- ▶ AI-use note if AI materially helped

34



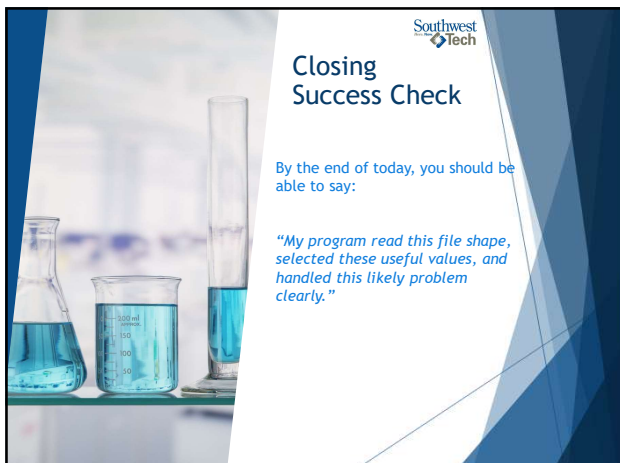
Southwest
Tech

A9 Structured Data Reader

- 1 Open provided file
- 2 Identify shape
- 3 Select useful values
- 4 Display readable result

Assignment 9

35



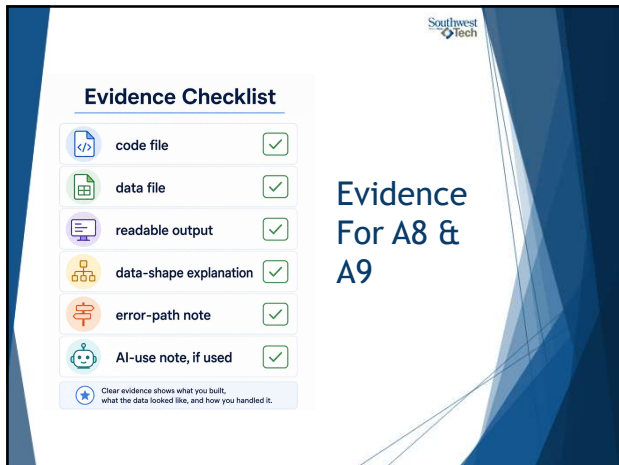
Southwest
Tech

Closing Success Check

By the end of today, you should be able to say:

"My program read this file shape, selected these useful values, and handled this likely problem clearly."

36



Southwest Tech

Evidence Checklist

-  code file ☒
-  data file ☒
-  readable output ☒
-  data-shape explanation ☒
-  error-path note ☒
-  AI-use note, if used ☒

 Clear evidence shows what you built, what the data looked like, and how you handled it.

Evidence For A8 & A9

37



Southwest Tech



Thank you!

Have Fun!

38
